

MIDTERM EXAM—SOLUTIONS

ECON301—ECONOMETRICS I

FEBRUARY 27, 2020

ALEJANDRO MELO PONCE

NAZARBAYEV UNIVERSITY

Instructions. The exam has a total of 100 points. I will grade it over 80. All questions have the same weight (20 points). Read the problems carefully; then decide which ones you want to attempt first. I suggest that you try to answer first the problems which you feel most confident about and then move on to the ones in which you don't feel so confident. I will provide you with paper in which to write your answers. Write your answers neatly and cross anything that you don't want me to consider. Please number each page you use. At the end of the exam I will staple all your papers. The duration of the exam is 80 minutes (1 hour and 20 minutes). You can use a calculator and your cheat sheet, but no other materials are allowed.

1. Let X and Y be any two random variables and suppose that their expectations exists. Prove the *conditional variance identity*:

$$\text{Var}(Y) = E(\text{Var}(Y|X)) + \text{Var}(E(Y|X))$$

Solution

$$\begin{aligned}\text{Var}(Y) &= E(Y^2) - (E(Y))^2 \\ &= E(E(Y^2|X)) - (E(Y|X))^2\end{aligned}$$

where we have used the *law of iterated expectations*; next using one of the identities for the variance, we have that

$$\begin{aligned}&= E(\text{Var}(Y|X) + (E(Y|X))^2) - (E(Y|X))^2 \\ &= E(\text{Var}(Y|X)) + (E(Y|X))^2 - (E(Y|X))^2\end{aligned}$$

but the last couple of terms are equal to $\text{Var}(E(Y|X))$ so that

$$= E(\text{Var}(Y|X) + \text{Var}(E(Y|X)))$$

2. Let X and Y be two continuous random variables whose joint distribution is

$$f_{X,Y}(x, y) = 24xy; \quad 0 \leq x \leq 1, 0 \leq y \leq 1 - x$$

- (a) Obtain the marginal distributions $f_X(x)$ and $f_Y(y)$.
- (b) Obtain the conditional distribution $f_{Y|X=x}(y)$.
- (c) Obtain the conditional expectation function $E(Y|X = x)$ for any value of x .

Solution

The domain of the random variables is

$$\mathcal{D} = \{(x, y) \in \mathbb{R}^2 : 0 \leq x \leq 1, 0 \leq y \leq 1 - x\} = \{(x, y) \in \mathbb{R}^2 : 0 \leq y \leq 1, 0 \leq x \leq 1 - y\}$$

We will need these two ways to rewrite this domain when computing the integrals below. Note that the pdf is $f_{X,Y}(x, y) = 24xy I_{\mathcal{D}}(x, y)$ where $I_{\mathcal{D}}(x, y)$ is the indicator function of $\mathcal{D}$.

$$(a) f_X(x) = \int_{-\infty}^{\infty} 24xy I_{\mathcal{D}}(x, y) dy = \int_0^{1-x} 24xy dy = 24x \frac{y^2}{2} \Big|_0^{1-x} = 12x(1-x)^2. \text{ Similarly, } f_Y(y) = \int_{-\infty}^{\infty} 24xy I_{\mathcal{D}}(x, y) dx = \int_0^{1-y} 24xy dx = 24y \frac{x^2}{2} \Big|_0^{1-y} = 12y(1-y)^2.$$

(b) The conditional distribution is

$$f_{Y|X=x}(y) = \frac{2xy}{x(1-x)^2} I_{[0, 1-x]}(y) = \frac{2y}{(1-x)^2} I_{[0, 1-x]}(y),$$

where the above is only valid when $x \neq 0$ or $x \neq 1$. To be very rigorous in here, you need to specify carefully that the conditional distribution is only valid when it respects the original domain $\mathcal{D}$. Conditional on $X = x$, this domain becomes $[0, 1-x]$. That is the reason for indicator function that appears in the display above. Note that the interval is open, since when x is either

$$(c) E(Y|X = x) = \int_0^{1-x} y \frac{2y}{(1-x)^2} dy = \frac{2}{(1-x)^2} \int_0^{1-x} y^2 dy = \frac{2}{(1-x)^2} \frac{y^3}{3} \Big|_0^{1-x} = \frac{2(1-x)^3}{3(1-x)^2} = \frac{2}{3}(1-x)$$

3. In one population, $E(Y_1) = \mu$ and $\text{Var}(Y_1) = \sigma_1^2$; in a second population, $E(Y_2) = \mu$ and $\text{Var}(Y_2) = \sigma_2^2$. The population variances are known, but their common expectation μ is unknown. Random samples of size n_1 and n_2 respectively are drawn from the two populations. The two samples are independent. It is proposed to combine the sample means $\bar{Y}_1$ and $\bar{Y}_2$ linearly into a single estimator of the common mean μ . Consider the class of estimators $\mathcal{W}$ that consists of all linear combinations of $\bar{Y}_1$ and $\bar{Y}_2$

$$\mathcal{W} = \{W_a = a_1 \bar{Y}_1 + a_2 \bar{Y}_2 : a_1 \text{ and } a_2 \text{ are constants}\}.$$

In what follows, let W_a be some estimator in $\mathcal{W}$.

- Find $E(W_a)$. What restriction on a_1 and a_2 is needed for W_a to be unbiased?
- Find $\text{Var}(W_a)$.
- Using the restriction that you found in (a), determine what values a_1 and a_2 must be for W_a to have the minimum variance among all estimators in the class $\mathcal{W}$. Call this the *minimum variance unbiased estimator* W_a^*
- Verify $\text{Var}(W_a^*)$ is indeed less than the variance of each of the two sample means.

Solution

- $E(W_a) = a_1 E(\bar{Y}_1) + a_2 E(\bar{Y}_2) = a_1 \mu + a_2 \mu = (a_1 + a_2) \mu$. Therefore, for unbiasedness we want $E(W_a) = \mu$, hence the restriction we must impose on the weights is $a_1 + a_2 = 1$.
- $\text{Var}(W_a) = a_1^2 \text{Var}(\bar{Y}_1) + a_2^2 \text{Var}(\bar{Y}_2) = a_1^2 \frac{\sigma_1^2}{n_1} + a_2^2 \frac{\sigma_2^2}{n_2}$.
- For W_a to have minimum variance, we must solve the following problem

$$W_a^* = \min_{a_1, a_2} \left\{ a_1^2 \frac{\sigma_1^2}{n_1} + a_2^2 \frac{\sigma_2^2}{n_2} \right\} \quad (1)$$

subject to $a_1 + a_2 = 1$.

There are many ways to solve problem (1). You can use techniques that you have learned in Intermediate Microeconomics and Multivariate calculus (optimization under constraints using the Lagrangian) but the method I will write here involves transforming the problem into a single-variable minimization, so that you can use single-variable calculus to solve it. Use the constraint to write $a_2 = 1 - a_1$ and plug this in the objective to obtain the problem

$$W_a^* = \min_{a_1 \in [0, 1]} \left\{ a_1^2 \frac{\sigma_1^2}{n_1} + (1 - a_1)^2 \frac{\sigma_2^2}{n_2} \right\} \quad (2)$$

Note that the objective in (2) is globally concave, since the second derivative with respect to a_1 is always positive^a for every $a_1 \in [0, 1]$. Thus, the first order condition is not only necessary *but also*

sufficient to identify a global minimum. The first order condition is

$$\begin{aligned} 2a_1^* \frac{\sigma_1^2}{n_1} - 2(1 - a_1^*) \frac{\sigma_2^2}{n_2} &= 0, \\ \Rightarrow a_1^* \left(\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2} \right) &= \frac{\sigma_2^2}{n_2}, \\ \Rightarrow a_1^* &= \frac{n_1 \sigma_2^2}{n_2 \sigma_1^2 + n_1 \sigma_2^2}. \end{aligned}$$

Thus, we also have that

$$a_2^* = 1 - a_1^* = \frac{n_2 \sigma_1^2}{n_2 \sigma_1^2 + n_1 \sigma_2^2}.$$

The minimum variance is:

$$W_a^* = \left(\frac{n_1 \sigma_2^2}{n_2 \sigma_1^2 + n_1 \sigma_2^2} \right)^2 \frac{\sigma_1^2}{n_1} + \left(\frac{n_2 \sigma_1^2}{n_2 \sigma_1^2 + n_1 \sigma_2^2} \right)^2 \frac{\sigma_2^2}{n_2} = \frac{\sigma_1^2 \sigma_2^2}{n_2 \sigma_1^2 + n_1 \sigma_2^2}$$

(d) It is easy to check, using straightforward algebra, that indeed we have

$$\frac{\sigma_1^2 \sigma_2^2}{n_2 \sigma_1^2 + n_1 \sigma_2^2} < \min \left(\frac{\sigma_1^2}{n_1}, \frac{\sigma_2^2}{n_2} \right).$$

^aThe second derivative is $2(\sigma_1^2/n_1) + 2(\sigma_2^2/n_2) > 0$ for all a_1 .

4. Consider the following population model

$$Y = \theta X^2 + \epsilon$$

where X is an explanatory variable and ϵ is the disturbance term. Suppose that the following two assumptions are true: (i) $E(\epsilon|X) = 0$ and (ii) $\text{Var}(\epsilon|x) = \sigma^2$. Suppose that you have access to a random sample of size n of observations (y_i, x_i)

- Show that $E(\epsilon|X) = 0$ implies the following condition: $E(\epsilon X^2) = 0$. *Hint: $E(\epsilon|X) = 0$ implies that $E(\epsilon X^2) = 0$.*
- Using either the method of moments or by minimizing the sum of squared residuals, derive the least squares estimator $\hat{\theta}$ of θ .
- Compute $E(\hat{\theta})$ and $\text{Var}(\hat{\theta})$.

Solution

- $E(\epsilon X^2) = E(E(\epsilon X^2|X)) = E(X^2 E(\epsilon|X)) = E(X^2 \cdot 0) = 0$, where the first equality follows from the law of iterated expectations and the third equality follows from the assumption that $E(\epsilon|X) = 0$.
- You can either use the moment condition $E(\epsilon X^2) = 0$ or minimize the sum of squared residuals, I will use the second method. The problem is

$$\min_b \sum_{i=1}^n (y_i - b x_i^2) \quad (3)$$

Notice that the objective in (3) is globally convex, since the second derivative with respect to b is positive^a for all b . Thus, the first order condition is not only necessary but also sufficient to identify a global minimum. The first order condition is

$$-2 \sum_i x_i^2 (y_i - \hat{\theta} x_i^2) = 0 \Rightarrow \sum_i x_i^2 y_i = \hat{\theta} \sum_i x_i^4 \Rightarrow \hat{\theta} = \frac{\sum_i x_i^2 y_i}{\sum_i x_i^4}$$

(c) Using the fact that for every observation we have that $y_i = \theta x_i^2 + \epsilon_i$ write

$$\hat{\theta} = \frac{\sum_i x_i^2 (\theta x_i^2 + \epsilon_i)}{\sum_i x_i^4} = \theta + \frac{\sum_i x_i^2 \epsilon_i}{\sum_i x_i^4}$$

Then, we have that

$$\begin{aligned} E(\hat{\theta} | x_1, \dots, x_n) &= \theta + \frac{1}{\sum_i x_i^4} \sum_i E(x_i^2 \epsilon_i | x_1, \dots, x_n) \\ &= \theta + \frac{1}{\sum_i x_i^4} \sum_i x_i^4 E(\epsilon_i | x_1, \dots, x_n) \\ &= \theta + \frac{1}{\sum_i x_i^4} \sum_i x_i^4 E(\epsilon_i | x_i) \\ &= \theta, \end{aligned}$$

where the first equality comes from the properties of expected value, the second one comes from conditioning on the sample $x_1, \dots, x_n$, the third one comes from the fact that in a random sample each observation is independent, and the fourth one comes from the assumption that $E(\epsilon | X) = 0$ for all values of X . Finally, using the law of iterated expectations, we conclude that $E(\hat{\theta}) = \theta$.

Similarly,

$$\begin{aligned} \text{Var}(\hat{\theta} | x_1, \dots, x_n) &= \text{Var}\left(\frac{\sum_i x_i^2 \epsilon_i}{\sum_i x_i^4} \mid x_1, \dots, x_n\right) \\ &= \frac{1}{(\sum_i x_i^4)^2} \text{Var}\left(\sum_i x_i^2 \epsilon_i \mid x_1, \dots, x_n\right) \\ &= \frac{1}{(\sum_i x_i^4)^2} \sum_i x_i^4 \text{Var}(\epsilon_i | x_1, \dots, x_n) \\ &= \frac{1}{(\sum_i x_i^4)^2} \sum_i x_i^4 \text{Var}(\epsilon_i | x_i) \\ &= \frac{1}{(\sum_i x_i^4)^2} \sum_i x_i^4 \sigma^2 \\ &= \frac{\sigma^2}{\sum_i x_i^4} \end{aligned}$$

Where the first and second equalities come from the properties of the variance, the third equality comes from the fact we are conditioning on $x_1, \dots, x_n$, the fourth equality comes from the independence across observations because of random sampling, and the fifth equality comes from the homoskedasticity assumption made in the beginning of the problem.

^aThe second derivative with respect to b is $2 \sum_i x_i^4 > 0$

5. Suppose that a researcher, using wage data on 250 randomly selected male workers and 280 female workers, estimates the OLS regression

$$\widehat{Wage} = 12.52 + \frac{2.12}{(0.23)} Male, \quad R^2 = 0.06, RSE = 4.2,$$

where *Wage* is measured in dollars per hour and *Male* is a binary variable that is equal to 1 if the person is male and 0 if the person is a female. Define the wage gender gap as the difference in mean earnings between men and women.

- Interpret β_1 : show that it is the gender wage gap and indicate how much this is estimated to be.
- What would happen if instead you specify the model as: $\log(wage) = \phi_0 + \phi_1 male + u$? What is now the interpretation of the parameter ϕ_1 ?
- Irrespective of the specification of the model, do you think that $E(u | male) = 0$ is true? Give reasons for your answer.

Solution

- (a) The model is $wage = \beta_0 + \beta_1 + u$. For this calculation, assume that $E(u|male) = 0$. Then

$$E(wage|male = 1) = \beta_0 + \beta_1$$

$$E(wage|male = 0) = E(wage|female) = \beta_0$$

Thus, $E(wage|male) - E(wage|female) = \beta_1$. This means that β_1 is the gender wage gap, which in this case is estimated to be 2.12 dollars per hour.

- (b) For this model, assuming we can hold all other factors in u constant, we have that

$$lwage_m = \log(wage|male) = \phi_0 + \phi_1 + u$$

$$lwage_f = \log(wage|female) = \phi_0 + u$$

So that $lwage_m - lwage_f = \phi_1$. Thus, the change in $\log(wage)$ between males and females is ϕ_1 . If you recall what saw in class, the change in the log approximates a *percentage change* in salary earned between genders if we multiply the coefficient by 100, i.e., $\% \frac{(wage_m - wage_f)}{wage_f} = 100\phi_1$.

- (c) The answer to this question depends on what do we think u includes, i.e., other factors affecting wage, and whether these factors change systematically with gender, i.e., the variable *male*. It is clear that u includes ability, intelligence, and innate skills and these factors are independent of gender. However, u also includes education, experience and many other sociological variables. Given the pervasive presence of gender discrimination and sexism across the world, it is highly likely that these sociological factors are correlated with *male*, i.e., they vary systematically with gender. Thus, it is highly unlikely that $E(u|male) = 0$ will hold. However, I must add that this also depends on the type of population we are taking our data from. In countries in which sexism is minimal (like in Northern Europe), perhaps we would have more room to argue that the sociological factors in u are really independent of gender. But even if this is the case, our sample must be really representative of the population: if our sample includes an overrepresented share of any of the genders, we might still have, within our sample, a systematic variation of the factors in u with respect to gender, and therefore, a failure of $E(u|male) = 0$.

All of the above motivates the inclusion of a richer model in which we control for more variables.